The Blockchain and its Many Uses

The potential for increased security, efficiency, and transparency that blockchain offers is too significant to ignore. It is evolving and disrupting traditional systems, proving that it has real industry altering powers.



Blockchain is a disruptive technology that digitises, decentralises, secures and incentivises the validation of transactions across industries. This technology has the potential to improve security, processes and systems in every domain where accurate, tamper-proof record-keeping is essential.

Trust: Blockchain is a decentralised, peer-to-peer network that records transactions to blocks, which are written to the distributed ledger based on consensus between all the participants in the network. The ledger is distributed to all the nodes and is not centralised. These factors foster trust among the participants.

Transparency: The data recorded on the blockchain ensures all the participants in the network have the distributed ledger copy; no party can maliciously change the data in the ledger, as the transaction is committed to the blockchain once all parties come to a consensus (based on applicable consensus algorithm).

Elimination of intermediaries: Blockchain technology eliminates the need for intermediaries. Once blockchain technology is adopted, consensus algorithms transparently record and verify transactions, eliminating the need for manual verifications by any third party. Cryptography helps in eliminating third-party intermediaries as keepers of trust.

Automation: With transactions executing almost immediately and significantly reducing the processing times (sometimes from days to hours), old time-consuming processes are replaced by automated verification and validation.

Key characteristics of blockchain technology

- ***Digital:*** Digitises all the information on the blockchain, thus eliminating the need for manual documentation.
- ***Distributed:*** Blockchain distributes control among all peers in the transaction chain, creating a shared infrastructure within an enterprise system. Participants independently validate information without centralised authority. There is no single point of failure because of distributed system operation. Even if one node fails, the remaining nodes continue to operate, ensuring no disruption.
- ***Immutable:*** All the transactions are immutable in the blockchain. Encryption is done for every transaction covering time, date, participants, and hash to the previous block.